Fiscal Year:	FY 2009	**Task Last Updated:**	FY 01/27/2010
PI Name:	Braby, Leslie Ph.D.		
Project Title:	Methods for Real Time Measurement of Dose and Charged Particle Spectrum		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	labraby@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979.862.1798
Organization Name:	Texas Engineering Experiment Station		
PI Address 1:	3133 TAMU		
PI Address 2:			
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2009 per PI (1/10) NOTE: Received NCE to 9/1/2007 per PI (6/07)		
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNJ04HH02G		
Performance Goal No.:			
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Task Description:	Low pressure, tissue equivalent, proportional counters are routinely used to measure the dose and radiation quality in space. A new approach to collecting and analyzing data will be developed in order to provide the information needed to improved models of cancer risk and to better evaluate radiation shielding. By measuring lineal energy simultaneously in 3 different size sites we expect to be able to distinguish between high energy heavy ions and lower energy lighter ions that have the same stopping power.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Results of simulations of energy deposition by 1 GeV/n iron ions in volumes of different diameter within a homogeneous absorber show that, for each site traversed by the iron ion, there is a very large number of small sites within range of the delta rays. Even though only a small fraction of those sites are actually hit by a delta ray, many more are hit by delta rays than by primary ions. The energy deposition events produced by iron ions form a small peak above 10 keV/µm. The most distinctive change in the energy deposition distribution with decreasing site size is the shift of this peak to lower values energy deposited. If there were no energy transfer by delta rays the maximum energy imparted by the primary ions would be equal to the stopping power (L) times the maximum chord length of the detector and for a spherical detector, we expect the maximum in the f(y) distribution to occur at 1.5 L. However, energy is carried away by delta rays so we expect the maximum of f(y) to occur at somewhat lower values of y, and for the maximum to shift lower as the site diameter decreases. The calculated maximum values of f(y) for 1 GeV/n iron ions occur at 193, 166, and 135 keV/µm for 2.5, 0.5, and 0.1 µm diameter sites, respectively. The stopping power of 1 GeV/n iron ions is 150 keV/µm, so if there was no energy carried away by delta rays we would expect the peak in f(y) to be at 225 keV/µm. Thus f(y) is actually at 85.8%, 73.8%, and 60% of the value based on L. The remainder of the energy, 14.2%, 26.2%, and 40%, is deposited by delta rays in a large number of low energy deposition events at significant distances from the primary ion track. Those delta ray events make up the majority of the energy deposition events in the case of 1 GeV/n iron ions, their being 33, 36, and 39 times as many delta ray events as primary ion events in 2.5, 0.5, and 0.1 micrometer diameter sites respectively. Since delta ray electrons are easily scattered by atomic electrons an equilibrium distribution of electron energies is established and f(y) for delta rays is nearly independent of the velocity of the ion that created them. These delta ray energy deposition events are similar to events produced by secondary electrons from low energy x-ray irradiation. Single delta ray interactions typically produce y < 10 keV/µm. However, a few sites close to the primary ion track may experience multiple delta ray tracks which constitute a single energy deposition event (because they were produced by the same primary ion) and may have y significantly greater than the maximum value for a single electron track. The probability of multiple delta ray tracks increases as the distance from the primary track decreases and increases with decreasing distance between delta ray tracks, i.e., increasing L. Energy deposited in a uniform medium, as described above, is fundamental to the biological effects of irradiation, but experimental measurements generally require a solid walled detector. The Monte Carlo simulations of energy deposition in a solid-walled detector show the changes expected as a result of the wall effect. Relatively few (compared to a homogeneous medium) delta ray events occur in the gas cavity because there are relatively few primary ion tracks through the wall within the range of the delta rays. When a primary track does occur just outside the gas cavity, a large number of delta rays are produced near the cavity (due to the high density of the wall) and there is a significant chance that two or more enter the cavity. Primary ions that cross the cavity are often accompanied by delta rays that were produced in the wall but enter the same cavity as the primary instead of a different small sites. These delta rays entering the site partly compensate for the delta rays leaving the site, and cause the y value to shift toward the stopping power of the primary ion. This is consistent with observations that f(y) measured by the Space Shuttle TEPC (tissue equivalent proportional counter) approximates the calculated f(L). Finally, the change in the probability density with simulated site size is much more pronounced for the solid walled detector than for the uniform homogeneous medium. The number of delta ray events per primary ion event increases rapidly because the cross sectional area of the wall that is within delta ray range of the detector cavity is proportional to the circumference of the sphere but the cross section for primary ion events is proportional to the detector area. The relatively rapid change with site size in the fraction of events due to primary ions for solid-walled detectors suggests that solid walled detector data will be better for evaluating the fraction of the dose that is due to high velocity particles. As a result, the experimental verification of these results was set up to utilize two solid walled detectors simulating sites with a 5 fold difference in simulated diameter. In order to evaluate the change in lineal energy distributions for mixed radiation fields, simulation results for a mixture consisting of 87% protons 12 % He ions, .33% oxygen ions, .33% silicon ions, and .34% iron ions were constructed from the data calculated for the individual ions. This mixture is less complex than the actual cosmic ray spectrum, but gives a reasonable estimate of the probability density distributions that would be measured by solid walled detectors simulating different site sizes in space. The proton and helium ion events dominate f(y), and completely mask the increase in delta ray events produced by heavy ions as the site size decreases. In an experimental measurement the stochastic variation in the number of primary ion events due to light ions would make it impossible to detect the increase in the number of delta ray events with decreasing site size. However, for the solid walled detector the relative change in the fraction of the events due to primary ions may be more useful. As the site diameter increases the energy loss due to delta ray transport decreases and the mean value of y for the primary ion events increases. This change is easily detected for charged particle accelerator beams but the magnitude of the change may not be sufficient to characterize the average velocity of the ions in the cosmic ray spectrum. Due to unexplained problems with detector operation in the pulsed beam at the NASA Space Radiation Laboratory it was not possible to provide experimental validation of the calculated results for delta ray event frequency. Even though these detectors accurately measured neutron spectra from a radioactive source it appears that the large detector had inconsistent gain and the small detector was producing extra pulses when exposed to the pulsed beam, but the mechanisms causing these errors could not be identified. Although it appears that the change in the spectrum with site size may be a practical way to determine the contribution of low energy neutrons to the dose in a mixed field or to determine the range of delta rays produced by high LET (linear energy transfer) radiation, it is apparently not feasible to determine the fraction of the cosmic ray dose delivered by delta rays because of the very large number of small events produced by protons and helium ions.
Bibliography Type:	Description: (Last Updated: 01/21/2010)
Articles in Peer-reviewed Journals	Wang X, Liu H, Xia Z, Braby LA. "Lineal energy as a function of site size for HZE radiation." Radiat Prot Dosimetry. 2006;122(1-4):367-8. PubMed PMID: 17251253 , Sep-2006

Articles in Peer-reviewed Journals	Braby LA, Ford JR. "Energy deposition in different diameter sites can help characterize the incident radiation." Radiation Measurements 2006 Oct-Nov;41(9-10):1051-250. http://dx.doi.org/10.1016/j.radmeas.2006.05.028 , Oct-2006